

Order Lepidoptera continued

Family Pieridae

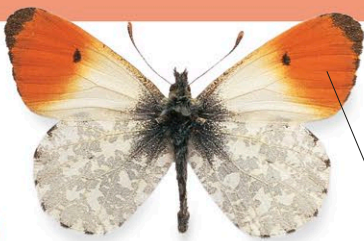
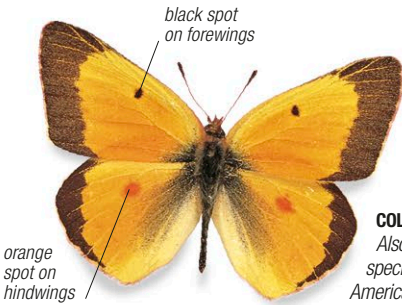
Whites and sulphurs

Occurrence 1,200 spp. worldwide; anywhere, often in groups at bird droppings, urine, or puddles in sunshine



The wings of these very common butterflies are usually white, yellow, or orange, with black or gray markings. The wingspan is $\frac{3}{4}$ – $2\frac{3}{4}$ in (2–7 cm) and the pigments in the wing scales are derived from by-products of the food eaten by caterpillars. Eggs are laid singly

or in groups of 20–100 on host plants. The pupae, which have a distinctive, spiny projection on the head, are held upright on the host plant by a silk belt.



ANTHOCHARIS CARDAMINES

The orange tip inhabits damp, grassy meadows and open woodland in Europe and temperate Asia. The undersides of the hindwings are dappled pale green.

COLIAS EURYTHEME

Also called the orange sulphur or the alfalfa sulphur, this species inhabits grassy, open areas and cultivated land in North America. Caterpillars feed on clover and can be pests of alfalfa.



PIERIS NAPI

The green-veined white inhabits grassy, open areas and cultivated land in Europe, Asia, and North America.

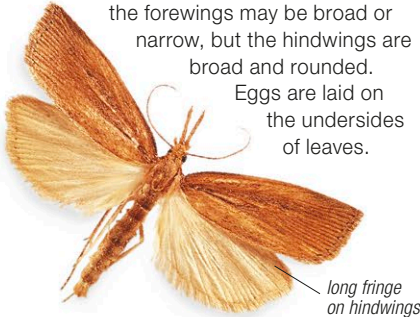
Family Crambidae

Grass moths

Occurrence 10,000 spp. worldwide, especially in warmer regions; on a range of plants, from aquatic plants to trees



Some species of grass moths have a short "snout" on the front of their head. Most are drably colored. Their wings span $\frac{3}{8}$ – $1\frac{3}{4}$ in (1–4.6 cm); the forewings may be broad or narrow, but the hindwings are broad and rounded. Eggs are laid on the undersides of leaves.



CHILO PHRAGMITELLA

The Wainscot veneer is found in reed beds, where its caterpillars feed on the stems of reeds belonging to the genus Phragmites.

Family Saturniidae

Saturniid moths

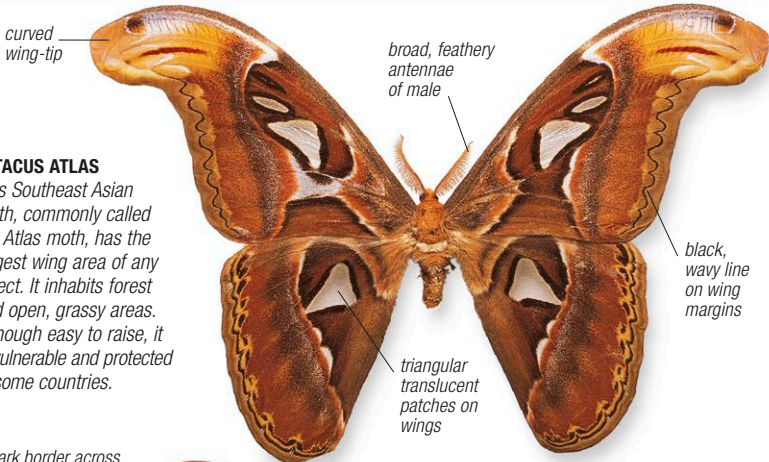
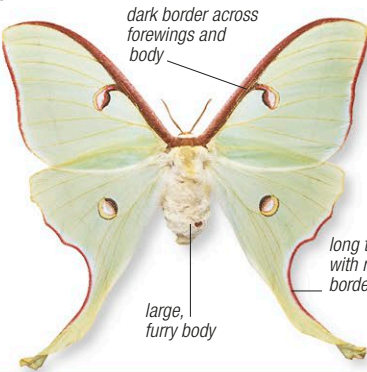
Occurrence 1,500 spp. worldwide, especially in wooded tropical and subtropical regions; on foliage



Variously known as emperor, moon, royal, and atlas moths, these are large and often spectacular insects with broad, often conspicuously marked wings. The wings span 2–12 in (5–30 cm), and the center of each may have an eyespot; some species have hindwing tails. Males have broad, feathery or comblike antennae. Fully grown caterpillars make dense cocoons, attached to the twigs of host plants, and feed on the foliage of a wide range of trees and shrubs. Adults do not feed.

ACTIAS LUNA

The American moon moth is found from Mexico to the Canadian border. The caterpillars feed on the leaves of deciduous trees.



ATTACUS ATLAS

This Southeast Asian moth, commonly called the Atlas moth, has the largest wing area of any insect. It inhabits forest and open, grassy areas. Although easy to raise, it is vulnerable and protected in some countries.

CALLOSAMIA PROMETHEA

The promethea moth inhabits woodland and cultivated areas of southern Canada and the USA. Females are heavy bodied and nocturnal; males are dark, day flying, and resemble Nymphalis antiopa (see p.571).



Family Sesiidae

Clear-winged moths

Occurrence 1,200 spp. worldwide, especially in northern temperate regions; around flowers or near host plants



These usually diurnal moths resemble wasps or bees; some even pretend to sting. They are black, bluish, or dark brown, with yellow or orange markings. The wings, large areas of which are transparent, span from $\frac{1}{2}$ – $1\frac{1}{2}$ in (1.4 to 4 cm) and produce a buzzing sound in flight. Many species are pests of fruit and other trees and shrubs.



SESIA APIFORMIS

With its transparent wings, the hornet moth, from Europe and Asia, convincingly mimics a hornet. Its caterpillars bore inside willows and poplars.

Family Sphingidae

Hawk moths

Occurrence 1,450 spp. worldwide, especially in tropical and subtropical regions; on foliage

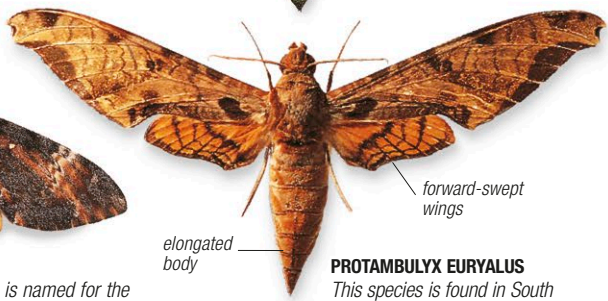
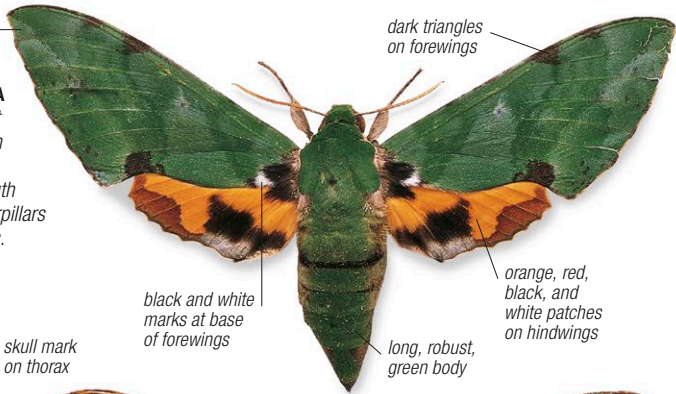
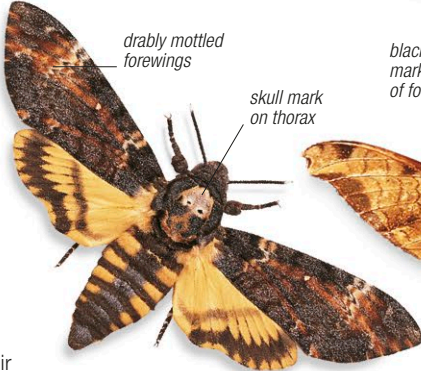


With their streamlined bodies and long, narrow wings, hawk moths are among the fastest flying insects, reaching speeds of over 30 mph (50 kph). Their wings measure $1\frac{1}{2}$ –6 in (3.5–15 cm) from tip to tip. Adult hawkmoths mostly suck nectar—some hovering like hummingbirds as they do this. They have a long, curled proboscis, up to 10 in (25 cm) long. Although some species are diurnal, most are active after dark. Hawkmoths often migrate long distances to breed. Their caterpillars are known as hornworms, because they have a spinelike process at the end of the abdomen. Some are very serious crop pests.

distinctive dark green forewings

EUCHLORON MEGAERA

Also called the verdant sphinx, this hawk moth inhabits forests and woodland of Africa south of the Sahara. Its caterpillars eat vines and creepers.



ACHERONTIA ATROPOS

The death's head hawk moth is named for the skull-like marking on its thorax. It is found in Africa and Asia, but also migrates to Europe.

PROTAMBULYX EURYALUS

This species is found in South America. Both sexes are very similar in appearance.